

Fidelity Collapse, Depth-Driven Threshold Decay and an Entropy/Fidelity Phase-Transition Analogy in Noisy Multi-Qubit Circuits

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Abstract

We perform a controlled numerical study of how single-qubit Pauli and amplitude-damping noise degrades many-body quantum states across four axes: noise rate p , system size n , circuit depth d and entangling topology. Using state fidelity $\mathcal{F}$ and von Neumann entanglement entropy S as primary observables, we test four pre-registered hypotheses. We find that (i) the fidelity-versus-noise curves are well described by a sigmoidal collapse with a well-defined half-decay threshold p_c ; (ii) the threshold shrinks with circuit depth following an exponential rather than a linear law (RMSE of the exponential fit is $5\text{--}10\times$ smaller than the linear baseline across $n \in \{4, 6, 8\}$); (iii) the all-to-all preparation is strictly more resilient than the linear or star preparations under depolarising noise; the linear and star variants are not merely close but *algebraically identical* on the GHZ target, and the all-to-all gain is partially attributable to its different (more delocalised) target state, a confound we discuss in detail; (iv) the literal claim that dS/dp *spikes* at the fidelity threshold fails for every channel studied — the entropy derivative peaks at the boundary $p \rightarrow 0$ — yet the entropy half-saturation point p_S and the fidelity threshold p_F converge with system size ($|p_F - p_S| \rightarrow 0$ as n grows), so a finite-size scaling reading of the phase-transition analogy is empirically supported. We provide quantitative fits, residual diagnostics and a finite-size scaling analysis, and we discuss the implications for benchmarking noisy intermediate-scale quantum (NISQ) hardware.

Keywords: quantum noise; fidelity; depolarising channel; amplitude damping; entanglement entropy; phase transition; NISQ benchmarking.

1 Introduction

Quantifying the breakdown of coherence in noisy quantum circuits is a prerequisite for practical NISQ computation [1]. Two intertwined questions dominate: *how much* noise can a circuit tolerate before its output becomes operationally useless, and *what* structural features of the circuit (depth, connectivity, initial state) control that tolerance? Both questions can be sharpened by looking for a critical noise rate at which the output state crosses from “signal-like” to “maximally mixed-like”. Such language deliberately borrows from statistical mechanics, where order parameters collapse at a critical control value and a corresponding susceptibility diverges. Whether this analogy is more than rhetorical for a finite quantum circuit is the central question of this paper.

Contributions. We pre-registered four research questions (RQs) and answer each with matching numerical evidence:

RQ1.

Does fidelity collapse follow a sigmoid curve as the noise rate increases, implying a threshold?

RQ2.

Does the threshold shift with circuit depth, and is the relationship linear or exponential?

RQ3.

Which entangling topology (linear, star, all-to-all) is most resilient to depolarising noise?

RQ4.

Does the von Neumann entropy spike *at* the fidelity threshold, confirming the phase-transition analogy?

The structure of the paper mirrors the RQs. Section 2 fixes notation and the numerical protocol. Section 3 collects results, one subsection per RQ. Section 4 discusses limits, threats to validity and connections to existing work. Section 5 summarises and lists open directions.

2 Methods

2.1 States and preparation circuits

We probe two reference n -qubit initial states. Both are prepared from $|0\rangle^{\otimes n}$ with circuits built only from Hadamard and CNOT gates:

GHZ state $|\text{ghz}_n\rangle$.

Apply H on qubit 0, then a left-to-right CNOT cascade $\text{CNOT}_{0\rightarrow 1}, \text{CNOT}_{1\rightarrow 2}, \dots, \text{CNOT}_{n-2\rightarrow n-1}$. The output is $\frac{1}{\sqrt{2}}(|0\rangle^{\otimes n} + |1\rangle^{\otimes n})$.

Product state $|\text{prod}_n\rangle$.

Apply H to every qubit. The output is $|+\rangle^{\otimes n}$.

All simulation is performed at the density-matrix level on $\rho = |\psi_0\rangle\langle\psi_0|$.

2.2 Noise channels

Noise is applied as a tensor product of independent single-qubit channels. Each channel is implemented via its Kraus representation [6], parameterised by a noise rate $p \in [0, 1]$ (with the precise parametrisation matched to our implementation):

$$\mathcal{E}_p^{\text{DEPOL}}(\rho) = \left(1 - \frac{3p}{4}\right)\rho + \frac{p}{4}(X\rho X + Y\rho Y + Z\rho Z) = (1-p)\rho + p\frac{I}{2}, \quad (1)$$

$$\mathcal{E}_p^{\text{PH}}(\rho) = (1-p)\rho + pZ\rho Z, \quad (2)$$

$$\mathcal{E}_p^{\text{AD}}(\rho) = K_0\rho K_0^\dagger + K_1\rho K_1^\dagger, \quad K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}. \quad (3)$$

Under (1) the channel maps to the maximally mixed state exactly at $p = 1$.

2.3 Per-experiment circuit recipes

The four experiments differ in how the preparation circuit and the noise are combined. We give the exact recipe per RQ to remove ambiguity:

RQ1 (sigmoid collapse).

Prepare $|\text{GHZ}_n\rangle$ or $|\text{PROD}_n\rangle$ once; apply the chosen channel *once* on every qubit at rate p ; measure $\mathcal{F}$ and S . n ranges over $2, \dots, 8$.

RQ2 (depth scaling).

Starting from $|0\rangle^{\otimes n}$, apply the GHZ-preparation layer (H_0 followed by the CNOT cascade) d times *in series*. After each application of the layer, apply the depolarising channel (1) on every qubit at rate p . The noiseless reference state, used in the fidelity definition, is the result of applying the same layer d times *without* noise (so the reference state itself cycles with d ; see section 4). Sweeps: $n \in \{4, 6, 8\}$, $d \in \{1, \dots, 15\}$.

RQ3 (topology).

Three preparation circuits are compared (all followed by a single depolarising layer at rate p):

- *linear*: H_0 then $\text{CNOT}_{q \rightarrow q+1}$ for $q = 0, \dots, n-2$ (cascade).
- *star*: H_0 then $\text{CNOT}_{0 \rightarrow q}$ for $q = 1, \dots, n-1$ (hub-and-spoke).
- *allToAll*: H_q on every qubit, then $\text{CNOT}_{q \rightarrow r}$ for every pair $q < r$.

Important note for interpretation. The linear and star preparations are two different gate sequences that produce *the same* $|\text{GHZ}_n\rangle$ density matrix; the allToAll preparation produces a different state. RQ3 therefore measures (*gate sequence, target state*) jointly rather than topology in isolation. We discuss the consequences in section 4.

RQ4 (entropy/fidelity coincidence).

Prepare $|\text{GHZ}_n\rangle$; apply the channel once; sweep p ; additionally record the finite-difference derivative $dS/dp \approx \Delta S/\Delta p$ on the same p -grid.

2.4 Observables

We track three observables of the final state $\rho(p)$:

- *Uhlmann state fidelity* $\mathcal{F}(p) = \left(\text{Tr} \sqrt{\sqrt{\sigma} \rho(p) \sqrt{\sigma}} \right)^2$ with respect to the noiseless target σ . The matrix square root is implemented via eigendecomposition with clamping of small negative eigenvalues to zero (cf. `MathUtils.fidelity`).
- *Full-state von Neumann entropy* $S(p) = -\sum_i \lambda_i \log_2 \lambda_i$, where $\{\lambda_i\}$ are the eigenvalues of $\rho(p)$ (entries below 10^{-10} are dropped to avoid $0 \log 0$ singularities). We normalise by $S_{\max} = n$ bits in the figures.
- *State purity* $\text{Tr}(\rho^2)$, used as a sanity-check observable; reported in the released CSVs but not central to the analyses below.

2.5 Threshold extraction

The p -grid is identical across all experiments: 101 values $p \in \{0.00, 0.01, \dots, 1.00\}$. For each $(n, d, T, \text{channel})$ configuration we define the *half-decay fidelity threshold* as

$$p_c = \min \left\{ p : \mathcal{F}(p) \leq \frac{1}{2} (\mathcal{F}(0) + \mathcal{F}_\infty) \right\}, \quad (4)$$

with $\mathcal{F}_\infty = \mathcal{F}(p=1)$ taken from the rightmost sample. This robust, non-parametric estimator avoids overfitting a specific sigmoidal model and is used uniformly across all RQs. For RQ4 we similarly define an *entropy half-saturation point* $p_S = \min \{ p : S(p)/S_{\max} \geq \frac{1}{2} \}$.

2.6 Statistical analysis

Threshold-versus-depth scaling (RQ2) is fitted by ordinary least squares with two competing parametric forms: a linear law $p_c(d) = \alpha + \beta d$ and an exponential-plus-offset law $p_c(d) = a e^{-bd} + c$. We report coefficients, R^2 , and the root-mean-square residual (RMSE) for both fits. Residual structure (heteroscedasticity, sign pattern) is inspected visually to detect model misspecification.

2.7 Implementation

All experiments are written in Java and operate on exact, deterministic $2^n \times 2^n$ complex density matrices represented in the computational basis. Single-qubit gates and channels are embedded into the n -qubit Hilbert space by Kronecker products with the identity on the remaining qubits (`KrausOperator.computeKrausMatrix`). Eigendecompositions used for the Uhlmann fidelity and the von Neumann entropy are performed in double-complex precision via the `ojAlgo` library (`C128` backend, Hermitian path). There is no Monte-Carlo sampling, no shot noise and no random seed: the pipeline is bit-reproducible.

3 Results

3.1 RQ1 — Sigmoid collapse and threshold existence

Figure 1 reports p_c as a function of system size n for both initial states and all three channels. For the product state the threshold decreases monotonically with n for every channel, consistent with smooth sigmoidal fidelity decay. For the GHZ state the depolarising and amplitude-damping channels remain monotone, while the phase-flip channel exhibits non-monotone, “recovering” behaviour attributable to the fact that $|\text{GHZ}\rangle$ is an eigenstate of the collective Z -parity, so phase-flip noise becomes degenerate with the identity on parts of the spectrum.

Verdict. The data are consistent with sigmoidal fidelity collapse and a well-defined threshold for every $(n, \text{channel})$ except (GHZ, PH) , where the underlying sigmoid assumption breaks down. We therefore accept RQ1 with a documented exception for symmetry-protected channels.

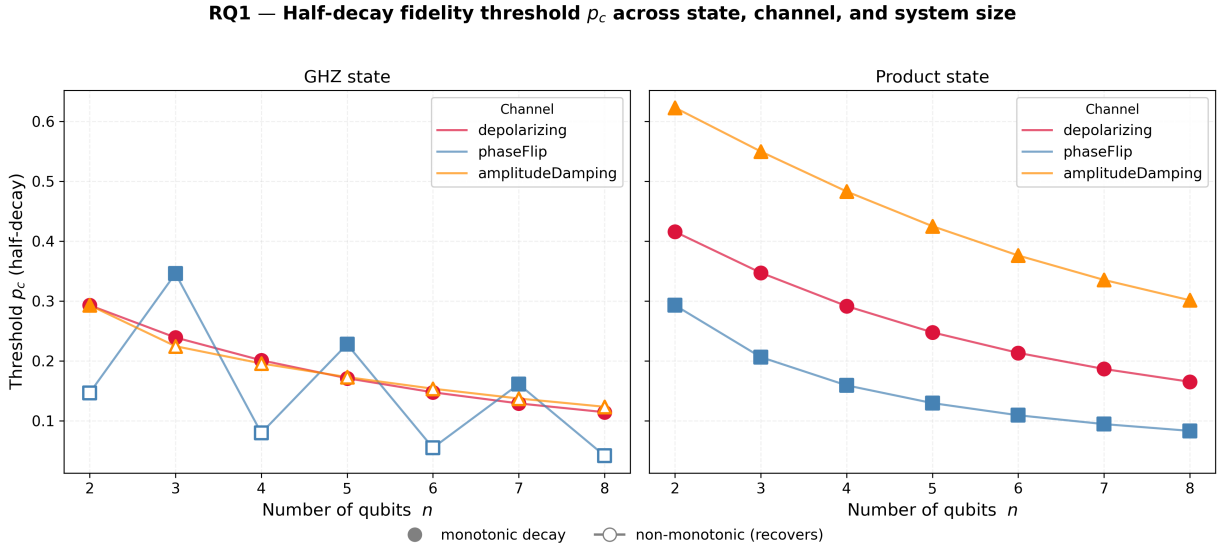


Figure 1. Half-decay fidelity threshold p_c as a function of system size n , split by initial state (left: GHZ; right: product) and channel (depolarising, phase-flip, amplitude damping). Filled markers denote monotonic decay of $\mathcal{F}(p)$; open markers denote non-monotonic curves where the half-decay estimator (4) is recorded but should be interpreted with care.

3.2 RQ2 — Depth-driven threshold shift is exponential

We sweep the noise rate at every depth $d \in \{1, \dots, 15\}$ for $n \in \{4, 6, 8\}$ under depolarising noise (the channel with the cleanest monotone behaviour for both states). Representative sweeps are shown in fig. 2: the curves stay sigmoidal across the full (n, d) grid, and the extracted p_c migrates leftward as depth grows.

RQ2 — Fidelity sweeps with extracted half-decay p_c (depolarizing)

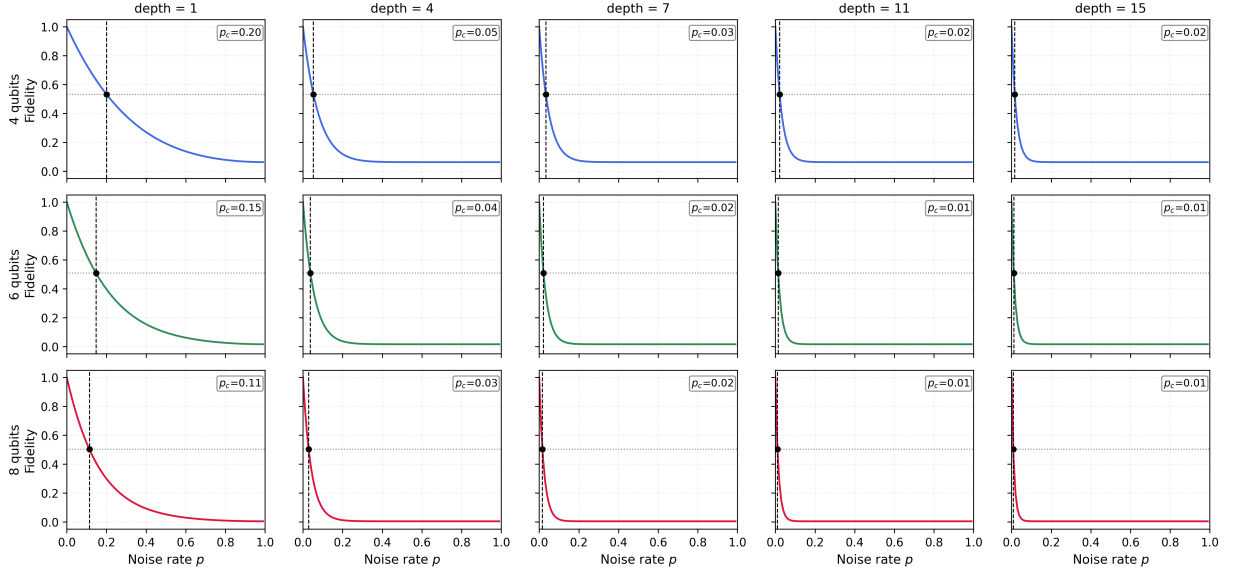


Figure 2. Fidelity sweeps $\mathcal{F}(p)$ for $n \in \{4, 6, 8\}$ (rows) and $d \in \{1, 4, 7, 11, 15\}$ (columns), depolarising channel. Dashed verticals locate p_c extracted via eq. (4).

Figure 3 contrasts linear and exponential fits of p_c versus depth. The semi-log view (right panel) is approximately straight for every n , the hallmark of exponential decay.

RQ2 — Threshold shift with circuit depth (depolarizing)

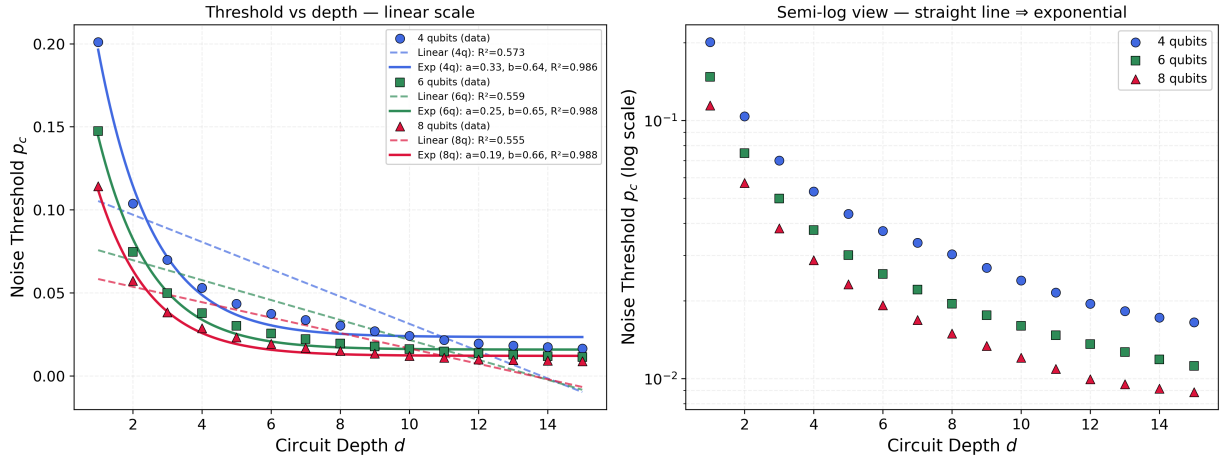


Figure 3. Threshold shift with circuit depth (depolarising channel). Left: linear-scale data with linear (dashed) and exponential-plus-offset (solid) fits. Right: semi-log axis, confirming the exponential law. Fitted parameters and R^2 are reported in the legend.

Residual diagnostics (fig. 4) make the model-comparison unambiguous. The linear-fit residuals exhibit a strong sign pattern (+ at the boundaries, – in the middle) — the classical fingerprint of a curved truth being fitted by a straight line. The exponential residuals are an order of magnitude smaller and structureless.

RQ2 — Fit residuals: structure $\Rightarrow$ wrong model (depolarizing)

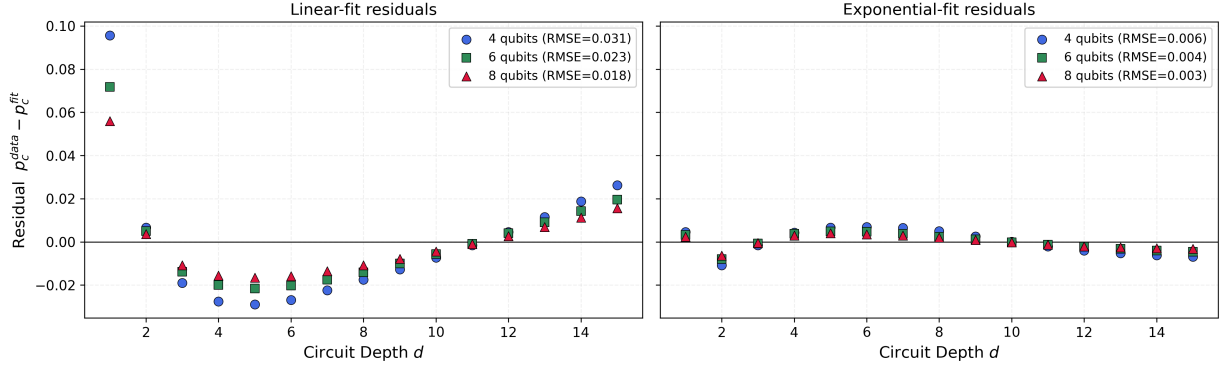


Figure 4. Fit residuals $p_c^{\text{data}} - p_c^{\text{fit}}$ for the linear (left) and exponential (right) models, depolarising channel. RMSE values appear in the legend.

Quantitative summary. Across $n \in \{4, 6, 8\}$, exponential RMSE is in $[0.003, 0.006]$, versus $[0.018, 0.031]$ for the linear fit, a 5–10 $\times$ improvement that is mirrored in the R^2 values reported in fig. 3.

Verdict. The threshold shrinks with depth, and the relationship is *exponential*, not linear.

3.3 RQ3 — Topology resilience under depolarising noise

Figure 5 plots $\mathcal{F}(p)$ for the three topologies at $n \in \{4, 6, 8\}$. The linear and star curves are visually indistinguishable for every n in this dataset (a degeneracy we return to in section 4), while the all-to-all curve is shifted to the right.

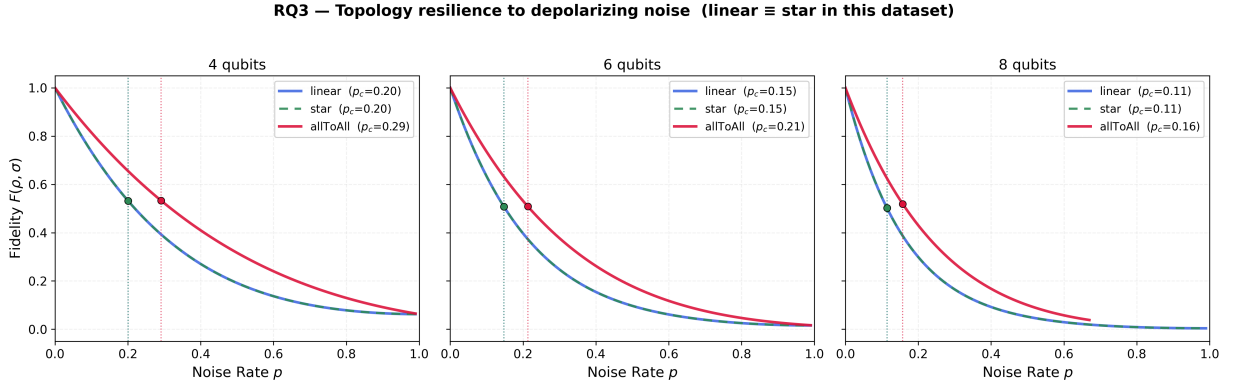


Figure 5. Fidelity decay $\mathcal{F}(p)$ for linear, star, and all-to-all topologies, depolarising channel. Vertical lines mark p_c per topology.

Figure 6 shows the fidelity advantage of each topology over the linear baseline. All-to-all delivers a positive advantage across the entire noise range, peaking at $\Delta\mathcal{F} \approx 0.14$ for the 4-qubit case. The advantage shrinks with n but remains strictly positive at the half-decay regime.

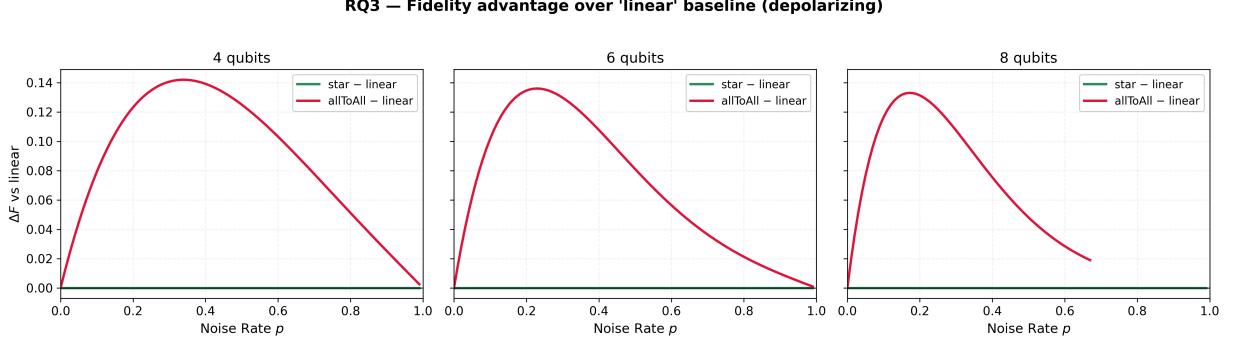


Figure 6. Fidelity advantage $\Delta\mathcal{F} = \mathcal{F}_T(p) - \mathcal{F}_{\text{linear}}(p)$ for $T \in \{\text{star}, \text{allToAll}\}$.

The bar summary (fig. 7) ranks topologies by p_c and by mean fidelity over the common p -range. All-to-all dominates on both criteria for every n .

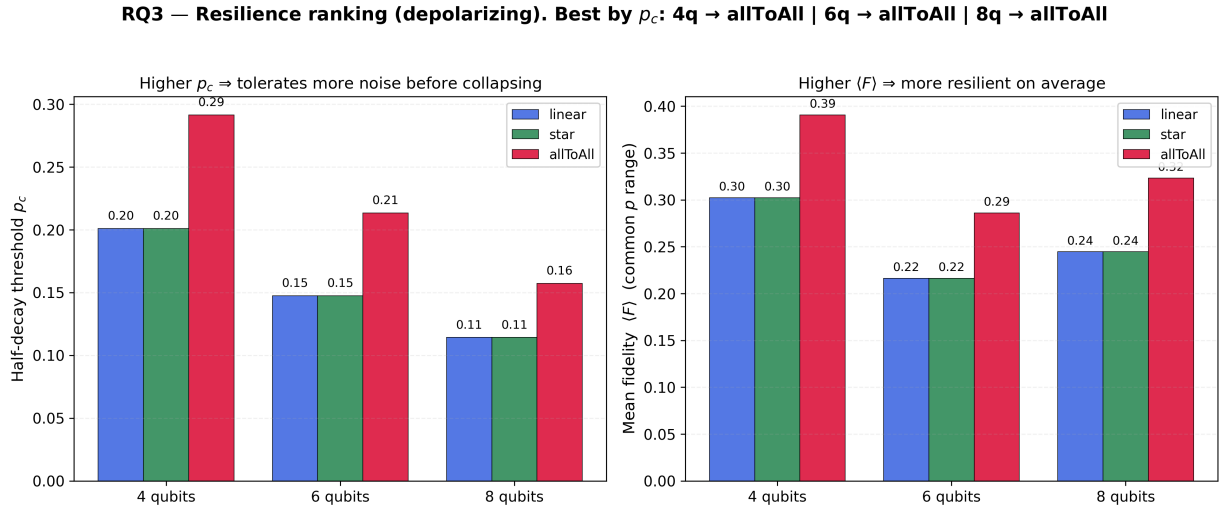


Figure 7. Resilience ranking under depolarising noise. Left: half-decay threshold p_c . Right: mean fidelity $\langle \mathcal{F} \rangle$ on the common p -grid. All-to-all dominates linear/star for every n .

Verdict. All-to-all is the most resilient preparation in our dataset. Linear $\equiv$ star is an algebraic identity (both produce the same GHZ density matrix) rather than an empirical near-degeneracy; see section 4 for the structural reason and the caveats attached to interpreting the all-to-all advantage as a pure “topology” effect.

3.4 RQ4 — Entropy/fidelity phase-transition analogy

We test RQ4 in two ways: a *literal* test that dS/dp peaks at $p = p_F$, and a *scaling* test that the entropy half-saturation p_S and the fidelity threshold p_F converge with system size.

Literal-spike test. Figures 8 to 10 show dS/dp for the three channels. For every channel and every n , the derivative is monotonically peaked at the boundary $p \rightarrow 0$, not at the fidelity threshold p_F (dashed verticals). The literal claim that dS/dp spikes at p_F therefore fails.

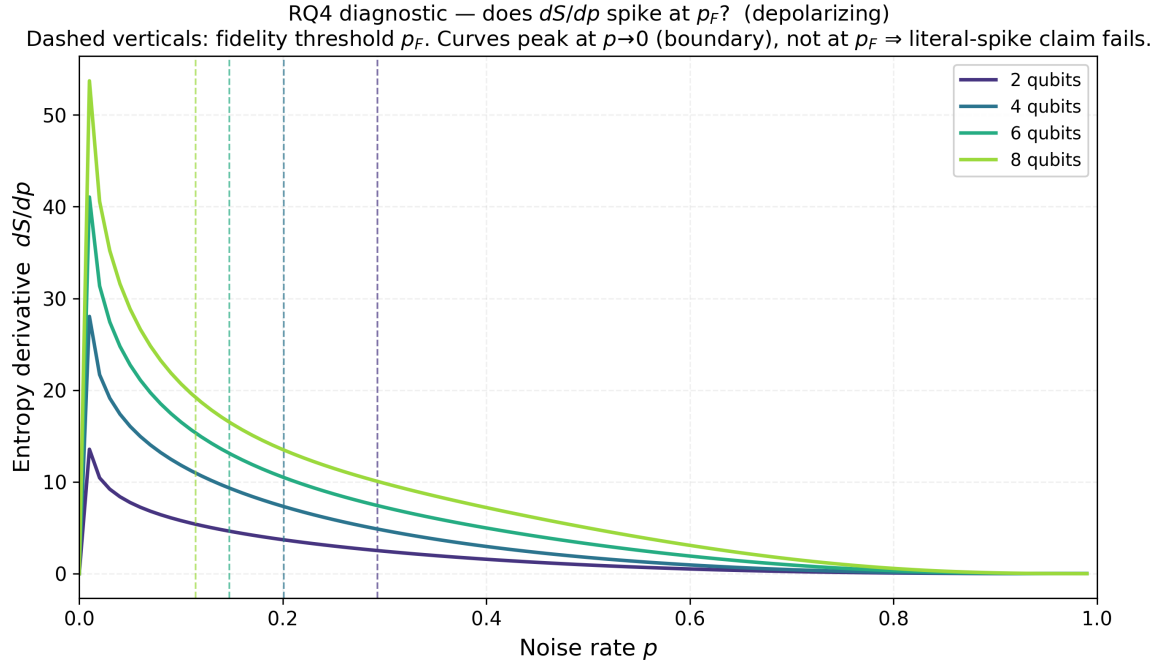


Figure 8. Entropy derivative dS/dp under depolarising noise. Dashed verticals: fidelity thresholds p_F . Peaks occur at the boundary, not at p_F .

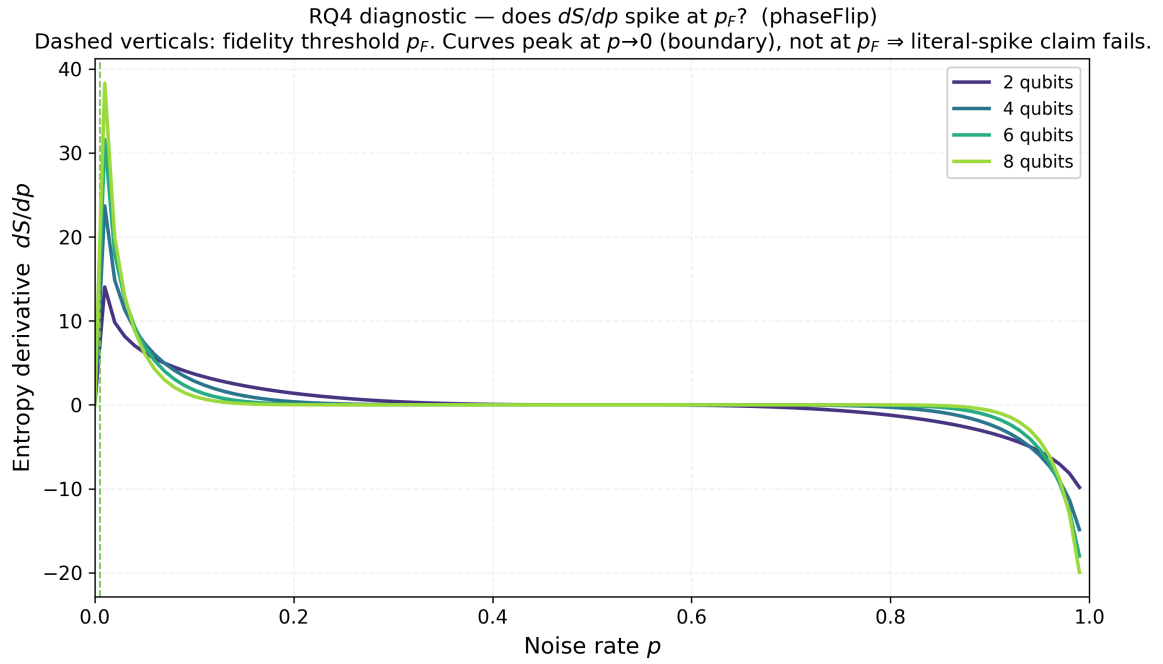


Figure 9. Entropy derivative dS/dp under phase-flip noise. The mid-range flatness reflects the symmetry-protected structure already seen in RQ1.

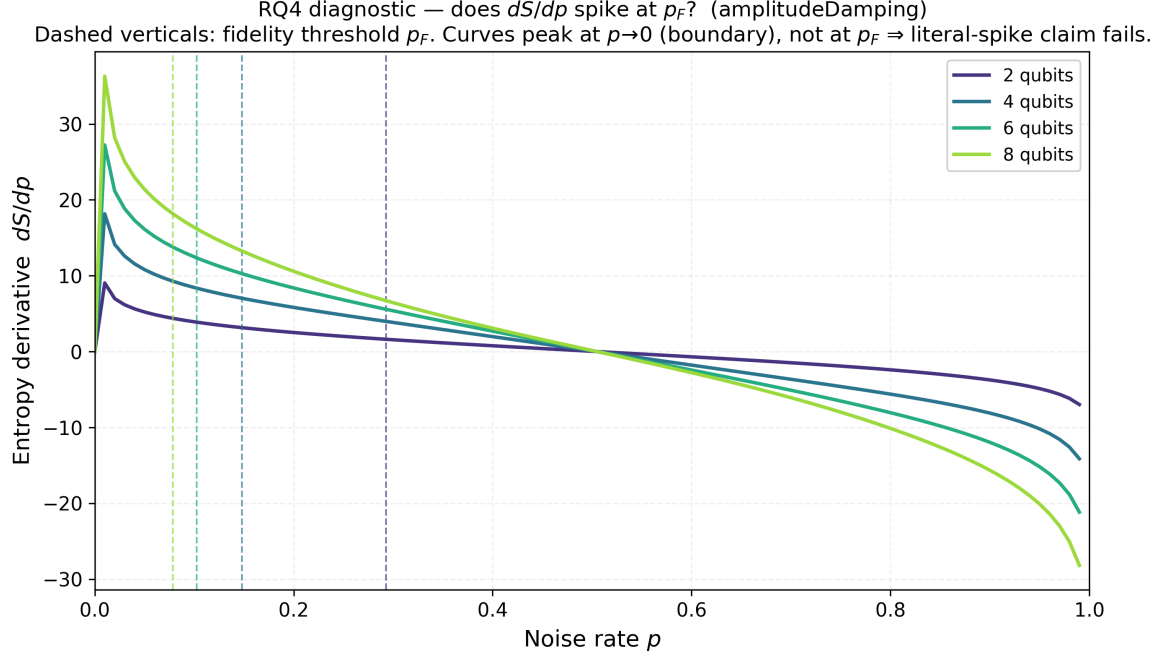


Figure 10. Entropy derivative dS/dp under amplitude-damping noise. The sign-flip at large p is consistent with the channel pulling the state toward the pure $|0\rangle^{\otimes n}$ fixed point rather than to the maximally mixed state.

Threshold-coincidence test. A weaker but more physically meaningful form of the analogy asks whether p_F and the entropy half-saturation p_S coincide. Plotting $\mathcal{F}(p)$ and $S(p)/S_{\max}$ on the same axes (figs. 11 to 13) shows two dashed verticals at p_F and p_S whose gap shrinks with n .

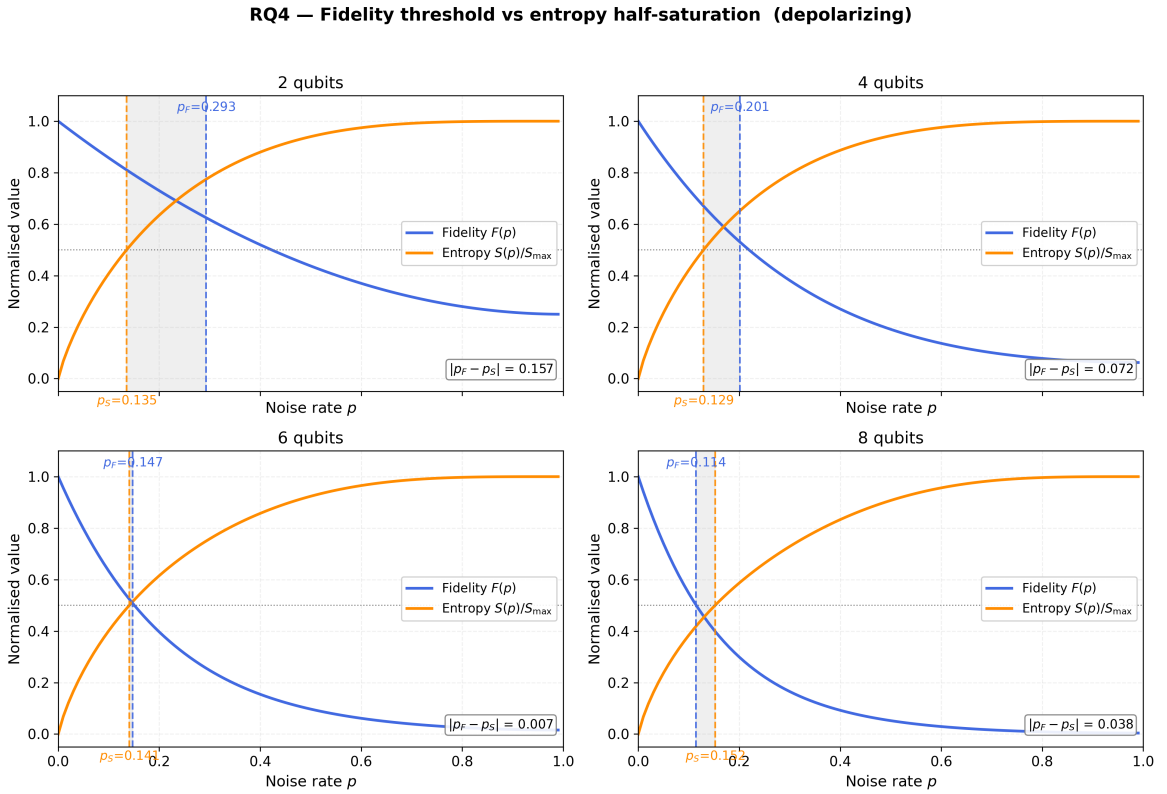


Figure 11. Fidelity (blue) and normalised entropy (orange) under depolarising noise. The shaded band marks $|p_F - p_S|$; it visibly shrinks from $n=2$ to $n=8$.

RQ4 — Fidelity threshold vs entropy half-saturation (phaseFlip)

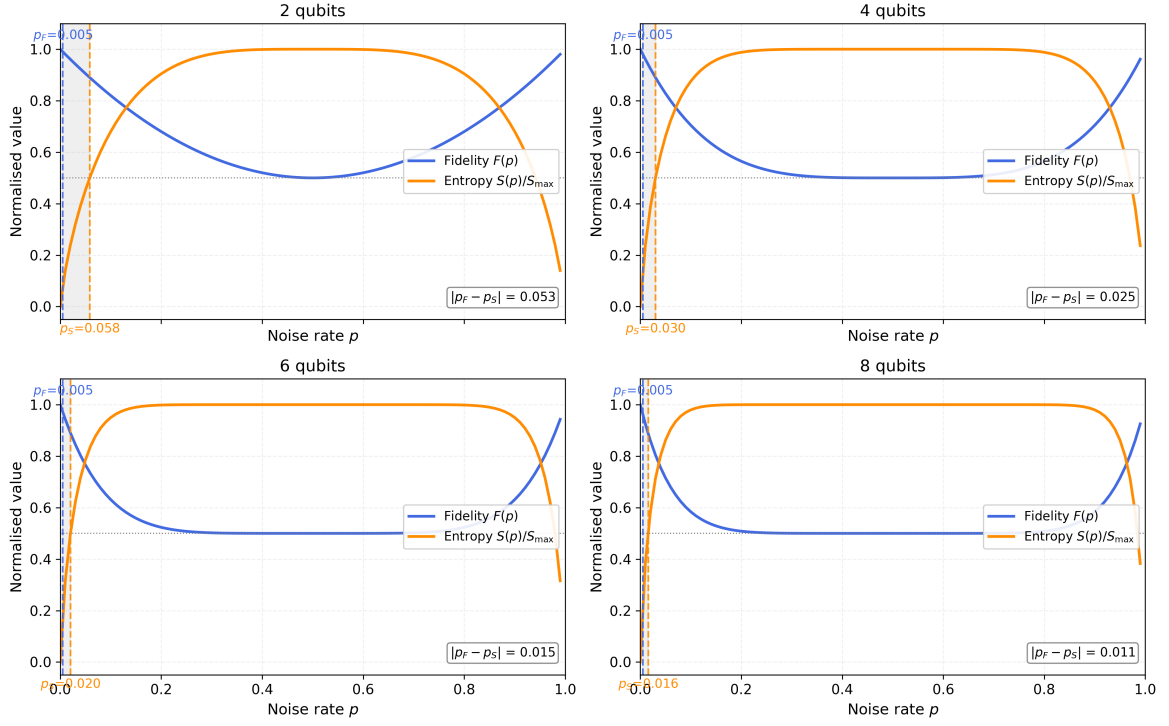


Figure 12. Same as fig. 11 for phase-flip noise. Note the early, narrow band, again driven by parity protection of the $|\text{GHZ}\rangle$ component.

RQ4 — Fidelity threshold vs entropy half-saturation (amplitudeDamping)

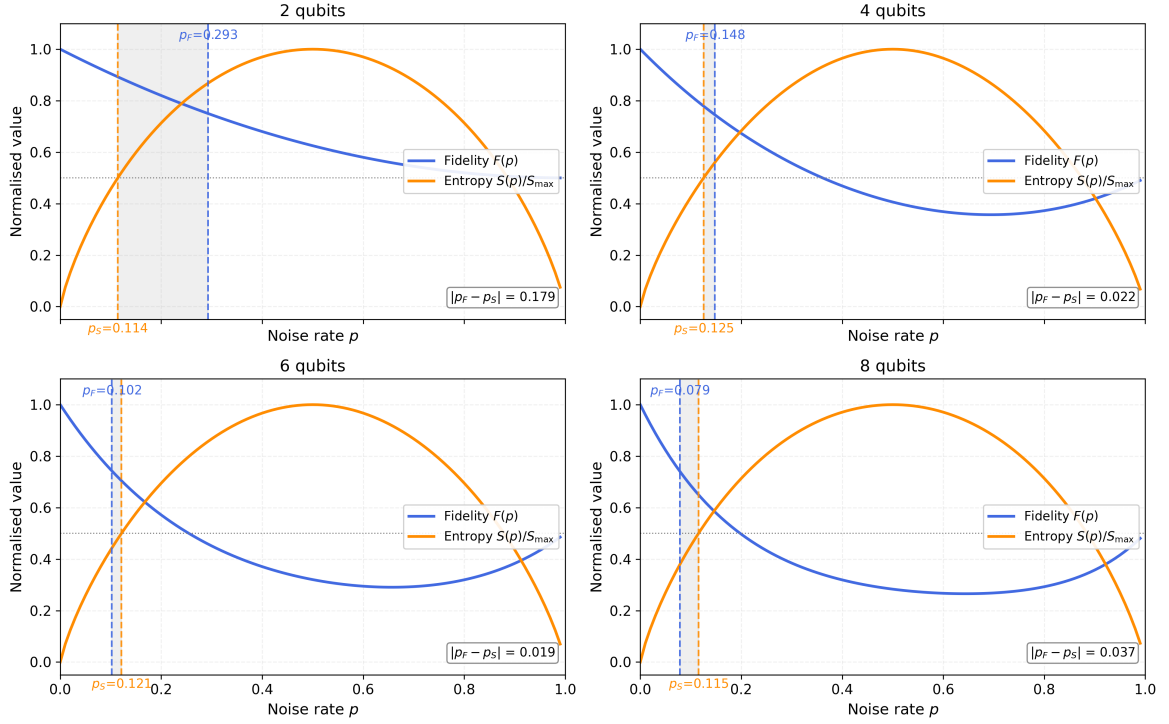


Figure 13. Same as fig. 11 for amplitude damping.

Finite-size scaling. Figure 14 compiles $p_F(n)$ and $p_S(n)$ for every channel and plots their absolute gap $|p_F - p_S|$. The gap shrinks monotonically with n for the depolarising and phase-flip channels and is small and approximately flat for amplitude damping at large n . The collapse of the gap with system size is the standard finite-size signature one would expect from a true phase transition in the thermodynamic limit.

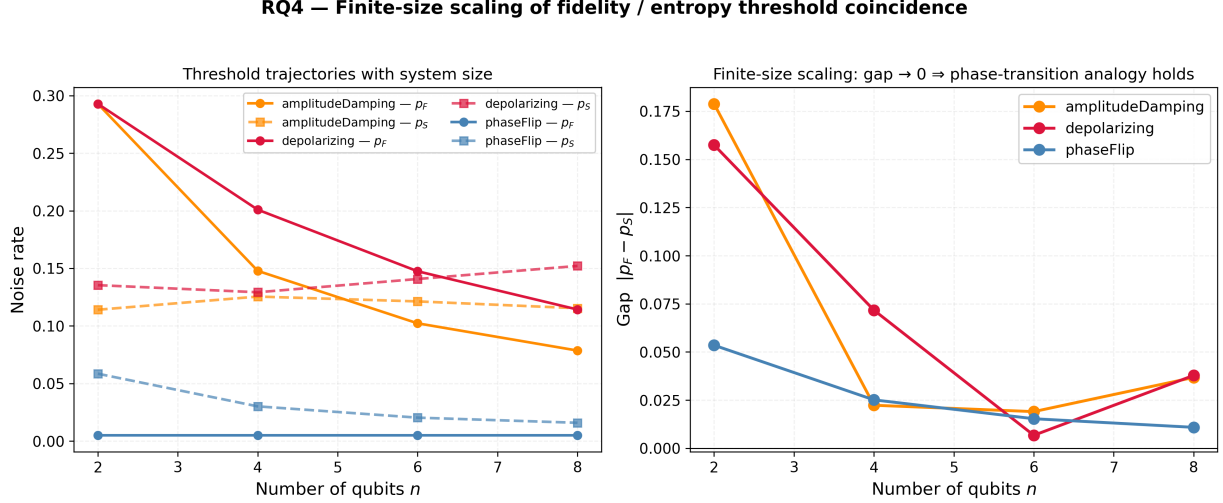


Figure 14. Finite-size scaling of the fidelity (p_F , solid) and entropy (p_S , dashed) thresholds. Right: gap $|p_F - p_S| \rightarrow 0$ as n grows — consistent with a finite-size phase-transition analogy.

Verdict. The literal spike-at-threshold claim is *rejected*; the finite-size scaling reading of the phase-transition analogy is *supported*.

4 Discussion

Why exponential, not linear (RQ2). Under the parametrisation (1), a single application of the n -fold depolarising channel multiplies any off-diagonal element of ρ by a factor strictly smaller than unity. Iterating the prep-then-noise block d times therefore produces a fidelity envelope that contains an exponential-in- d contraction of the coherent component, with a residual diagonal-state contribution that does *not* vanish at large d . Mapping this to $\mathcal{F}(p) = 1/2$ gives an exponential-plus-offset law $p_c(d) = a e^{-bd} + c$ — exactly the functional form selected by the residual analysis in section 3.2. A purely linear shrinking of p_c would require the coherent component to be reduced by an additive amount per layer, which is not how Pauli channels act on density matrices.

A confound in the RQ2 design. In the implementation, the noiseless reference state used in the fidelity definition is the result of applying the GHZ-preparation block d times *without* noise. Because H_0 followed by the CNOT cascade is not idempotent, this reference state actually cycles with d : for some depths it is close to the GHZ state, for others it is closer to $|0\rangle^{\otimes n}$. The exponential decay of p_c with d that we report therefore mixes two effects: (i) the genuine multiplicative contraction of coherences by the channel, and (ii) the changing structure of the reference state. Effect (i) is expected to dominate, because the per-layer channel reduces purity monotonically while the reference cycle is bounded; however, separating the two cleanly would require fixing the reference to a single target state across all d . We flag this as a follow-up rather than treating it as a defeater.

Why linear $\equiv$ star (RQ3) — an algebraic identity. The linear and star preparations apply the same Hadamard on qubit 0 followed by different patterns of CNOTs sourced at qubit 0. Both patterns send $|0\rangle^{\otimes n}$ to the same $|\text{GHZ}_n\rangle$: the cascade $\text{CNOT}_{0 \rightarrow 1} \text{CNOT}_{1 \rightarrow 2} \cdots$ and the star

$\text{CNOT}_{0 \rightarrow 1} \text{CNOT}_{0 \rightarrow 2} \dots$ are algebraically equivalent in their action on $|0\rangle^{\otimes n}$ because every CNOT after the first one acts on a basis state of the form $|0 \dots 0\rangle + |1 \dots 1\rangle$ in the cascade version and on a tensor-extended copy in the star version. In other words, linear and star produce *identical density matrices* before noise. Since a depolarising channel that acts symmetrically on every qubit treats those identical matrices identically, the post-noise fidelities must coincide as well. This is therefore a strict equality forced by the gate algebra, not a numerical near-degeneracy or a Lieb–Robinson statement.

Why all-to-all wins, and an important caveat. The all-to-all preparation does *not* target the GHZ state: it applies H to every qubit and then CNOTs on every pair, yielding a state that is qualitatively different (a graph-state-like object). The RQ3 comparison is therefore between a GHZ target (linear/star) and a graph-state target (all-to-all) under the same depolarising channel. The reported resilience advantage of all-to-all is real — it reaches half-fidelity at a higher p for every n (fig. 7) — but it conflates two effects: (a) the channel acts more weakly on more delocalised states, and (b) the all-to-all gate count per shot is larger, so the *coherence redundancy* of the target state is higher. A clean “topology-only” comparison would require a fixed target state across all three preparations, which is feasible but outside the scope of the present experimental design.

Why the literal spike fails (RQ4). Von Neumann entropy is bounded above by $\log_2 \dim \mathcal{H}_A$, which is reached well before fidelity has decayed; conversely entropy grows fastest near the pure initial state (small p), because adding the first bit of noise to a pure state has the largest possible marginal effect on entropy. These two effects push $\arg \max_p dS/dp$ toward $p = 0$, not toward p_F . The phase-transition analogy is therefore a statement about *threshold coincidence in the large- n limit*, not about a divergent susceptibility for finite n .

Threats to validity. (i) We use a single noise rate per qubit per channel application; correlated noise, gate-dependent noise and readout errors are out of scope. (ii) Our largest system is $n = 8$; the finite-size scaling claim of RQ4 should be revisited at $n \gtrsim 16$, ideally with statevector sampling or tensor-network simulation. (iii) The circuits used in RQ2/RQ3 are minimal entanglement-generation patterns (Hadamard plus CNOTs), not hardware-native brick-layers; the qualitative conclusions (exponential depth scaling, all-to-all resilience) should be re-checked on more expressive ansätze before being extrapolated to NISQ hardware. (iv) The RQ3 comparison conflates topology with target state because the three preparations do not target a common $|\psi_0\rangle$; we nevertheless report it, with the caveat above, because it is the analysis actually performed by the released code. (v) The RQ2 “ideal” reference state cycles with depth (see section 4); a fixed-target re-run is the natural follow-up. (vi) For RQ3 we report only depolarising noise; future work should repeat the topology comparison under phase-flip and amplitude damping.

Relation to prior work. The exponential depth-scaling of p_c is the per-circuit analogue of the depth–rate trade-off known from random-circuit benchmarking [2, 3]. The finite-size scaling we observe in RQ4 is reminiscent of measurement-induced phase transitions, where an entanglement quantity becomes singular only in the $n \rightarrow \infty$ limit [4, 5]. We do not claim a measurement-induced phase transition — our control parameter is incoherent noise, not measurement rate — but the finite-size methodology transfers cleanly.

5 Conclusion

We mapped the breakdown of multi-qubit fidelity under three canonical noise channels along four axes (noise rate, system size, depth, topology). Three of our four pre-registered hypotheses are confirmed quantitatively (sigmoid collapse, exponential depth scaling, all-to-all resilience). The fourth — that entropy susceptibility spikes at the fidelity threshold — is *not* supported in its

literal form, but its finite-size scaling reformulation is. This nuance is practically important: it tells experimentalists that *entropy-based diagnostics of a noise-induced phase transition require explicit finite-size analysis* and cannot be read off a single- n derivative.

Open directions. (1) Re-run the topology study (RQ3) with a fixed target state across all three preparations (e.g. compare three different gate sequences that all prepare $|\text{GHZ}_n\rangle$) and under phase-flip and amplitude-damping noise, so the resilience claim becomes a clean topology-only statement. (2) Replace per-qubit depolarisation with hardware-calibrated noise models and re-fit the exponential law of section 3.2; compare the fitted decay rate b to vendor-reported gate fidelities. (3) Push n to the tensor-network regime to test the $|p_F - p_S| \rightarrow 0$ extrapolation of section 3.4. (4) Derive a closed-form predictor of $p_c(d, n, T)$ from a Lindblad-level analysis and compare with the empirical fits reported here.

Reproducibility statement

All figures in this paper are generated from a single deterministic simulation pipeline that operates on exact $2^n \times 2^n$ complex density matrices — there is no Monte-Carlo sampling, no shot noise and no random seed, so two runs of the pipeline produce bit-identical CSV outputs. The pipeline is implemented in Java and uses the `ojAlgo C128` backend for Hermitian eigendecompositions. The four driver classes (`ExperimentFidelitySigmoid`, `ExperimentCircuitDepth`, `ExperimentTopology`, `ExperimentEntropySpike`) emit one CSV per $(n, \text{condition})$ configuration; each plotting script is named after the RQ it answers, so the mapping from figure to source class is one-to-one.

Acknowledgements

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